

A 1-Mb 28-nm 1T1MTJ STT-MRAM With Single-Cap Offset-Cancelled Sense Amplifier and *In Situ* Self-Write-Termination

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Abstract—1T1MTJ spin-transfer-torque (STT)-MRAM is a promising candidate for next-generation high-density embedded non-volatile memory. This paper presents a 1-Mb 28-nm 1T1MTJ STT-MRAM with improved sensing margin and reduced power consumption. An offset-cancelled sense amplifier is proposed, using only a single capacitor, to improve sensing margin and accelerate read speed. To save write power, an *in situ* write-self-termination method is proposed where the sense amplifier is reconfigured without area overhead to continuously monitor the write operation and shutoff the write drivers as soon as the magnetic transition occurs in the bitcell. A prototype chip achieves 2.8- and 3.6-ns read access time at 25 °C and 120 °C, respectively. The *in situ* write-self-termination scheme reduces write power by 47% and 60% with 20-ns write access time at 25 °C and 120 °C, respectively.

Index Terms—1T1MTJ, MRAM, offset cancellation, self-write-termination, sense amplifier, spin-transfer-torque (STT).

I. INTRODUCTION

EMBEDDED memories play a pivotal role in very large scale integration (VLSI) systems to support the increasing need of data storage in various applications such as many core systems, mobile systems, and Internet of Things (IoT) systems. Conventional 6T SRAM provides the highest performance; however, it suffers from a large area and high standby power [1], [2]. 1T1C DRAM significantly reduces cell area, but it requires refresh to sustain the data, inducing a considerable amount of power consumption. To lower standby power, non-volatile memories such as NOR flash can store data in a permanent way even when power is OFF. However, scaling below 28 nm is expensive in terms of cost for embedded NOR flash. Moreover, NOR flash consumes high energy during the program and erase because of its high-voltage operation and

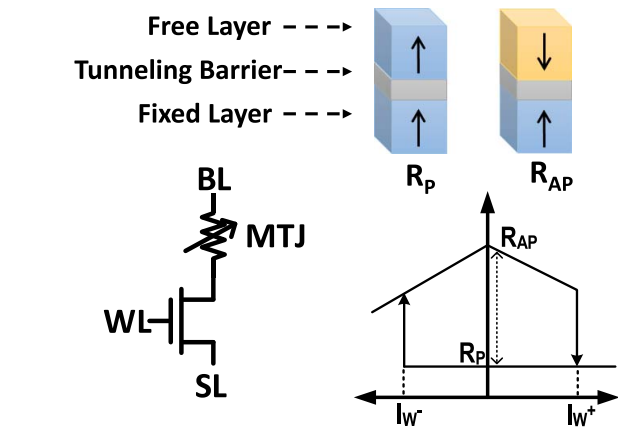


Fig. 1. Basic cell structure of 1T1MTJ STT-MRAM.

slow performance [3]. Endurance is also a limitation for NOR flash to applications requiring frequent write [4].

Recently, spin-transfer-torque (STT)-MRAM has attracted a lot of attention as a promising candidate for next-generation high-density embedded non-volatile memory [5]–[11]. STT-MRAM uses a magnetic tunnel junction (MTJ) to store the data. As shown in Fig. 1, the MTJ consists of one fixed layer, one tunneling barrier layer, and one free layer. If the current flows from free layer to fixed layer, the spin polarization of free layer will become same as fixed layer. Then, the parallel resistance R_p keeps low. If the current flows in the opposite direction, then the spin polarization of the free layer will be different, keeping the antiparallel resistance high as R_{AP} . The MTJ can be placed in backend processes, therefore providing compact area and good scalability. Compared with NOR flash, STT-MRAM has improved performance, higher endurance, and lower write energy [12], [13]. However, STT-MRAM still suffers from two challenges.

The first challenge is the limited read margin. Fig. 2 illustrates the distribution of R_p and R_{AP} under different conditions. Since the resistance difference between the two states is only about twice, the non-overlapping region is quite narrow. As a result, the sensing margin is limited, challenging sense amplifier design. Moreover, the temperature can shift

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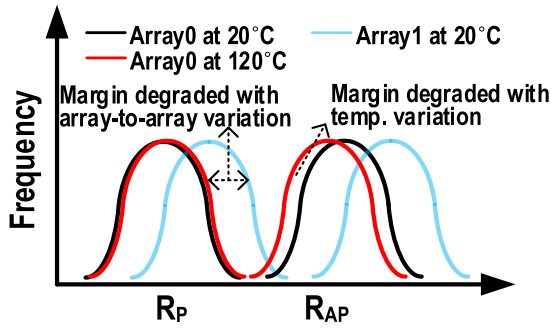


Fig. 2. Resistance distributions change with process and temperature variations, narrowing sensing margin.

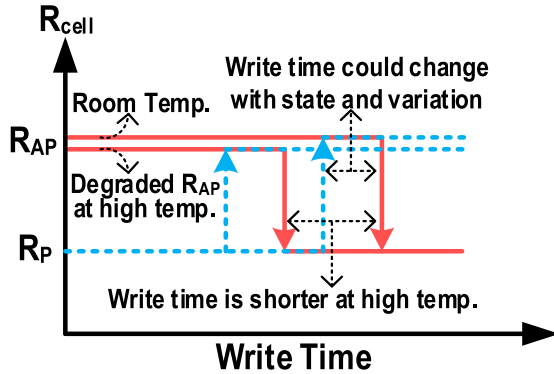


Fig. 3. Required write time varies with process and temperature variations, wasting write power consumption with fixed write pulselength.

the resistance distribution. When the temperature increases, R_P almost keeps the same, while R_{AP} degrades, making the resistance distributions even closer [14], [15]. Furthermore, resistance distributions of R_P and R_{AP} can shift under process variation, resulting in the requirement for a read reference that tracks process.

Another design challenge is the high instantaneous write power. Even though STT-MRAM can be written without high voltage, hundreds of microampere for tens of nanosecond are still needed to flip the MTJ in order to support the 10-year retention requirement for non-volatile memory applications. Fig. 3 shows that the required switching time from R_P to R_{AP} or R_{AP} to R_P could be unbalanced due to both the asymmetry of MTJ switching and transistor driving ability [16], [17]. Also, this unbalance changes from cell to cell under process variation. Moreover, high temperature accelerates the write speed. As shown in the red lines in Fig. 3, less write time is required to flip the cell at 120 °C compared to 20 °C. As a result, a fixed write time that ensures successful write for all conditions wastes a significant amount of energy at typical conditions.

In this paper, we propose an offset-cancelled sense amplifier, using only a single capacitor, to improve sensing margin and accelerate read speed. In addition, dual dummy columns are added in each array to generate read reference voltage tracking rowwise PVT variation. To save write power, an *in situ* write-self-termination method is proposed using reconfigured read sense amplifier to continuously detect the write completion and

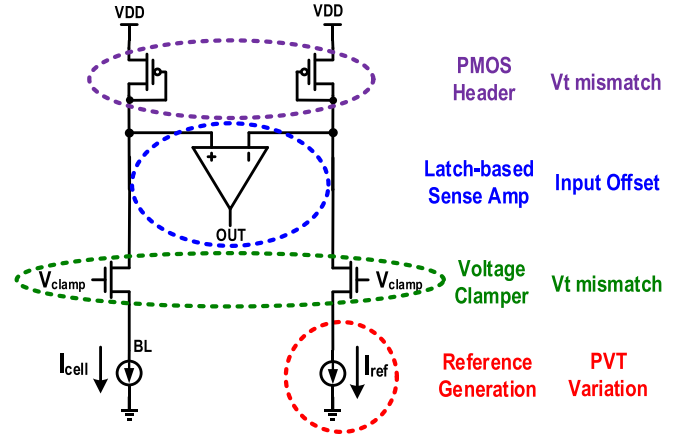


Fig. 4. Variation sources in conventional current sense amplifiers.

self-shutoff the write drivers without any timing penalty. The 1-Mb STT-MRAM was fabricated in TSMC 28-nm embedded MRAM technology, achieving 2.8- and 3.6-ns read access time at 25 °C and 120 °C, respectively. With *in situ* self-write-termination, the write power can be saved by 47% and 60% with 20-ns write access time at 25 °C and 120 °C, respectively [18].

The remainder of this paper is organized as follows. Section II discusses the general limitations of conventional sensing scheme. Section III describes the proposed self-reference generation and an offset-cancelled sense amplifier. Section IV presents the proposed *in situ* self-write-termination. Section V discusses the measurement results of the proposed design, and finally, conclusions are presented in Section VI.

II. CONVENTIONAL SENSING SCHEME

Fig. 4 shows a simplified diagram of a conventional sense amplifier for non-volatile memory. From a circuit point of view, it has four main variation sources, which degrades the sensing margin: 1) the first is the reference generation circuit that provides the reference current to compare with the cell current. Since the cell current varies with temperature and process variation, the generated reference current has to track PVT variation to alleviate their impact; 2) the second is the voltage clamper. NMOS-based voltage clamper is widely used to avoid unwanted write during a read operation. These NMOSs not only limit the voltage applied on the cell but also work as common gate amplifier to boost the voltage difference between the bitline (BL) and reference BL. However, the V_t mismatch between the two NMOSs induces offset to the sense amplifier, which could cancel out the advantage from the voltage amplification; 3) the third is from the PMOS header. PMOS header converts the current difference into voltage difference and sends it to the latch-based second-stage sense amplifier. The V_t mismatch between the two PMOS headers directly generates voltage offset to the sense amplifier; and 4) the last one is the innate offset of the latch-based sense amplifier induced by the mismatch between the inverter pair due to the process variation or the layout effect.

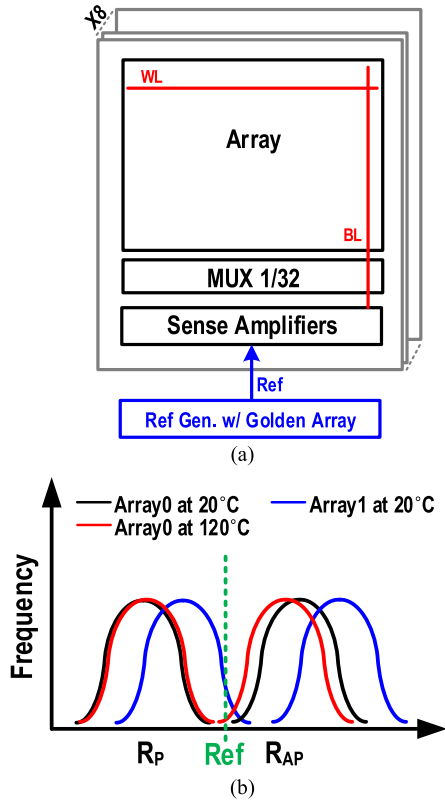


Fig. 5. (a) Conventional reference generation and (b) constant reference cannot well track PVT variation.

For the reference generation, the conventional method typically employs a small specific array with current mirrors to generate the required reference current [Fig. 5(a)]. This reference current is shared by all common arrays and calibrated according to global corners after fabrication. However, the shared fixed reference cannot finely track PVT variation for each array, affecting sensing margin as shown in Fig. 5(b).

For the offset on the latch-based second-stage amplifier, there have been many works done to minimize the mismatch between the inverter pair [19]–[23]. However, the common drawback of these offset-cancelled sense amplifiers is using the dual capacitors, which could induce over 30% area overhead for the sense amplifier.

For the offset induced by V_t variation of NMOS voltage clampers or PMOS headers, the conventional method usually increases the transistor size to minimize the mismatch in between, trading off area for robustness.

III. PROPOSED SENSING METHOD

A. Self-Generated Voltage Reference

In this design, we propose two circuit techniques in a sensing scheme that can alleviate all the four variation sources mentioned earlier. First is the self-generated voltage reference. Fig. 6 shows a detailed block diagram of the 1-Mb STT-MRAM, which contains eight arrays, each with 256 wordlines by 514 columns. To generate the reference voltage, each array has two extra columns. As shown in Fig. 7, the reference cell in one column is programmed to R_P ; while the

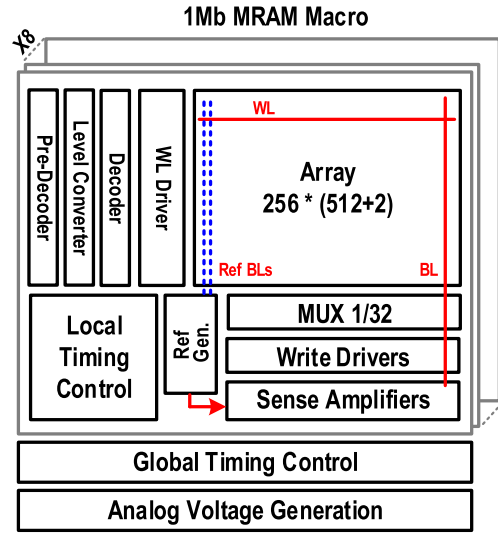


Fig. 6. Detailed diagram of the 1-Mb MRAM.

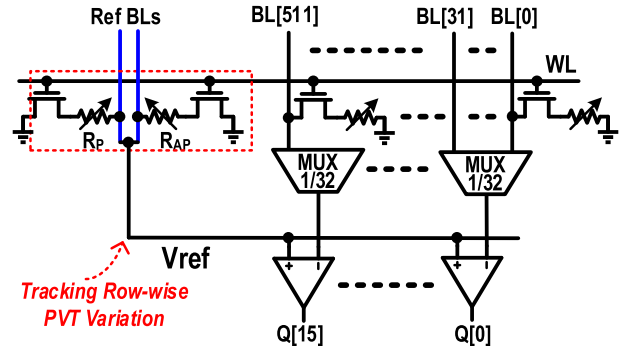


Fig. 7. Rowwise reference cells are used to generate read reference voltage tracking rowwise PVT variation.

other is R_{AP} . When WL is asserted, the two reference columns are activated. By setting the reference BL current to around twice of I_{read} , an intermediate reference voltage is generated and shared by 16 sense amplifiers. Since the reference cells are located in the same row as the selected cells, rowwise PVT variations are well tracked and compensated. Area overhead of the self-reference generation is less than 1% of the whole macro area.

B. Offset-Cancelled Sense Amplifier

Second is a constant large current-based fast voltage sensing scheme to improve both the read performance and the sensing margin. Fig. 8 shows the detailed read circuits. We apply a constant read current to the selected BL, which can be as high as 1/4 of write current. Since the current is high, it quickly develops sufficient BL voltage difference that can be easily sensed by the offset-cancelled sense amplifier.

To avoid disturbance, NMOS voltage clumper limits the voltage applied on the BL; while PMOS current clumper keeps constant current flow through the BL. Compared with PMOS current clumper, NMOS voltage clumper provides faster precharging speed when BL level is low. However, NMOS voltage clumper suffers from mismatch in between

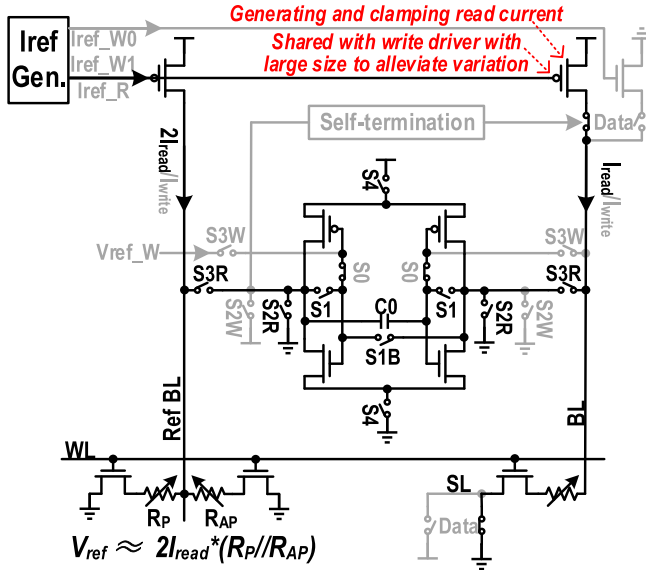


Fig. 8. Constant current-based voltage sensing with single-capacitor-based offset cancellation.

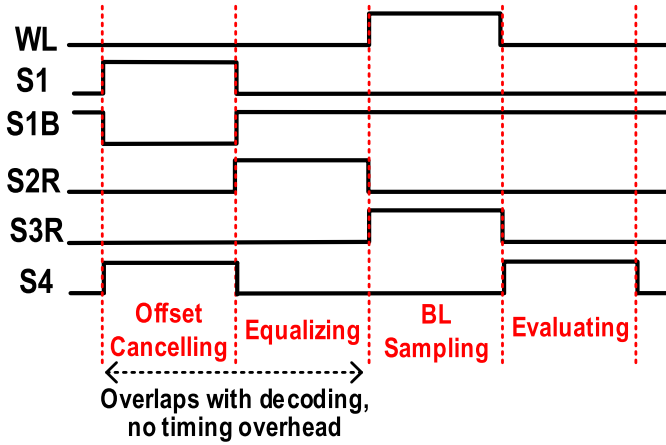


Fig. 9. Detailed read timing sequence.

when the size of them were limited. In our design, we use mismatch alleviated PMOS current clumper to limit the current on the BL. Since our PMOS current clumper can be shared with write drivers, their transistor size is less constrained. With the large size of the PMOS current clamping headers, the mismatch in between can be significantly alleviated without additional area overhead.

For the second-stage sense amplifier, a single-cap-based offset cancellation method is proposed to minimize mismatch.

Fig. 9 shows the detailed timing sequence of the read operation. A read cycle starts with an offset cancellation phase. During offset cancellation, the inputs of the inverters are connected to their outputs. As shown in Fig. 10, the capacitor C_0 samples the trip voltages of the two inverters V_L and V_R at its two sides. During equalizing, the left side of C_0 is discharged to ground. As a result, the right side stores $V_R - V_L$ that is the difference between the trip voltages of the two inverters. During BL sampling, this voltage $V_R - V_L$ is added to V_{ref} . The input of the right inverter becomes $V_{ref} + V_R - V_L$, cancelling

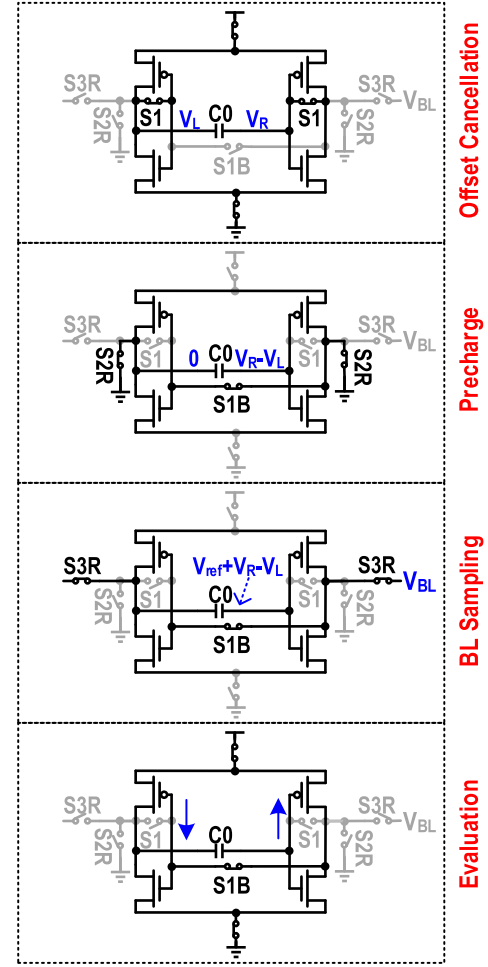


Fig. 10. Offset cancellation scheme using single cap.

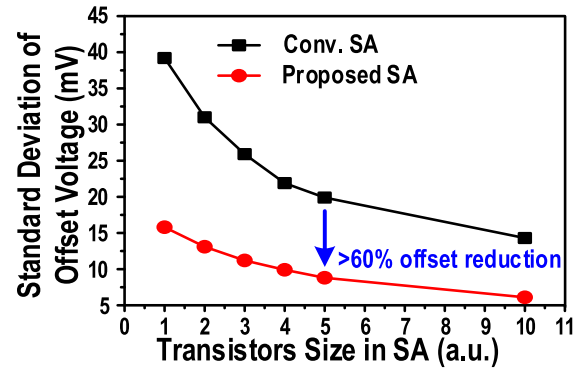


Fig. 11. Compared with conventional sense amplifier, input offset is reduced by >60%, significantly improving sensing margin.

out the mismatch. Since the mismatch of the two inverters is largely alleviated during evaluation such that the sensing speed can be significantly improved. Both offset cancellation and equalizing phases are performed simultaneously with address decoding, which can avoid any timing penalty for the proposed method.

Fig. 11 shows the simulated input offset across transistor sizing for both the conventional sense amplifier and the proposed sense amplifier with $1 \mu\text{m}^2$ MOS cap, which is

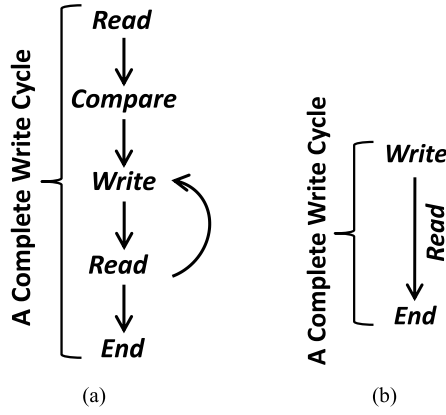


Fig. 12. (a) Conventional read-compare-write and write-verify-write to save write power [5]. (b) Proposed *in situ* self-termination write enables read and write at the same time.

large enough to provide voltage coupling. Based on 10-K Monte-Carlo simulations, the standard deviation of the input offset is reduced by over 60% compared to the conventional sense amplifier with the same transistor sizes. Using only a single capacitor, the area of the sense amplifier is reduced by approximately 15% compared with voltage sense amplifiers using dual capacitors for offset cancellation with similar reduced offset voltage [20], [21].

IV. PROPOSED IN SITU SELF-TERMINATION WRITE

A. Read-Before-Write and Write-Verify-Write

During a conventional write operation, hundreds of microampere current is continuously applied on the cell for tens of nanosecond even when the write is done, wasting power consumption. In [5], a combination of read-compare-write and write-verify-write was proposed to lower write power for STT-MRAM (Fig. 12). It starts with read operation at the beginning of write procedure and compare the input data with existing data. Then, a short write pulse will only be applied on the necessary bits that need to be written, avoiding redundant write. After that, the data are read out to check whether the write is done. If not, short write pulse will be applied again. The short write and verify loop will continue until all data are written successfully. This long sequence can help to reduce write power, but multiple cycles are required since read and write cannot happen at the same time, increasing write latency.

B. Proposed Write Operations

In this design, we proposed an *in situ* self-termination write scheme, which enables write and read at the same time. We continuously read during write to perform verify instead of multiple interrupted write-read-write steps, saving power without timing penalty.

During write, a constant current flows through the cell. When the free layer in the MTJ flips, the resistance of MTJ changes abruptly, which can be detected by observing the BL voltage. For write 1 (Fig. 13(a)), current flows from BL to SL and the resistance changes low from R_{AP} to R_P . As a result, the BL voltage drops when write 1 finishes. For write 0

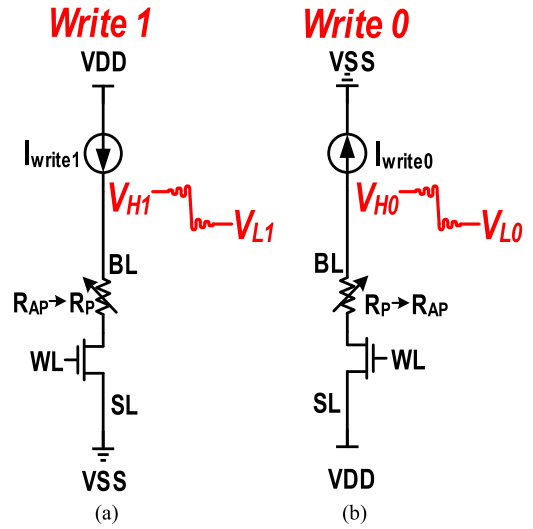


Fig. 13. BL voltage drops in both (a) write 1 and (b) write 0 cases.

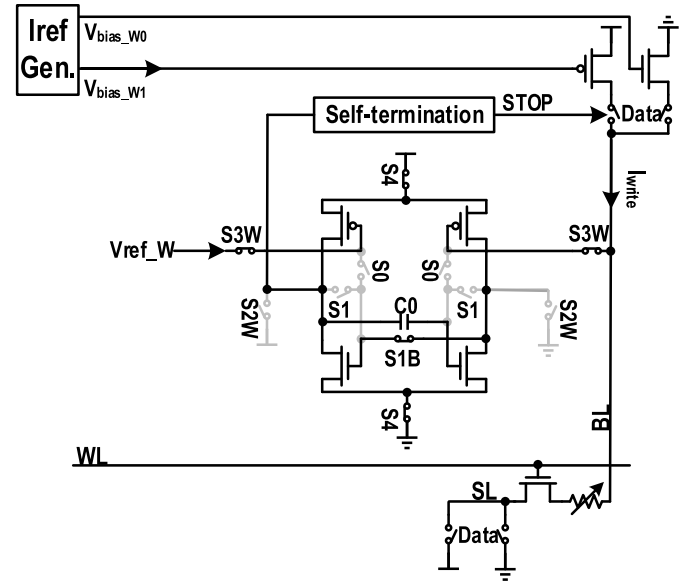


Fig. 14. Detailed write circuits. Read sense amplifier is reconfigured as a continuous gate-connected voltage sense amplifier to detect the BL voltage drop.

(Fig. 13 (b)), current flows from SL to BL and the resistance changes high from R_P to R_{AP} , decreasing BL voltage as well. Therefore, the BL voltage drops in both cases when the write completes, simplifying the detection for write completion. Write 1 and write 0 have different reference voltages, which can be selected by voltage MUX according to input data.

Fig. 14 shows the detailed circuits for the proposed write operation. The read sense amplifier is reconfigured as a continuous gate-connected voltage sense amplifier to detect the BL voltage drop. Since the write driver and the detection circuit are shared with the read circuit, area overhead is negligible.

Fig. 15 shows the detailed timing sequence of the proposed *in situ* self-termination write scheme. Same as read, a write cycle also starts with an offset cancellation phase, during which the inputs of the inverters are connected to their outputs. The capacitor C_0 samples the trip voltages of the

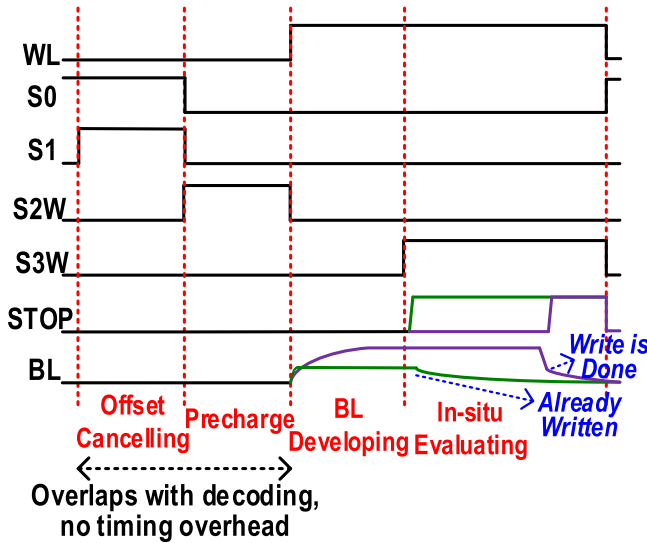


Fig. 15. Detailed write timing sequence.

two inverters at its two sides. Then, the left side is precharged to VDD, while the right side is discharged to ground. And cap C0 stores the mismatch voltage of the two inverters. Again, offset cancellation and precharge phases are overlapped with decoding, avoiding timing penalties. Next, the WL turns on for a while to stabilize the BL voltage. At this step, the sense amplifier is isolated with BL to avoid glitch. After the BL voltage is stable, the sense amplifier connects to the BL in order to continuously track the voltage change. If the cell stores the same data with the input, BL voltage stays low and write completion is detected immediately. The write current will be cutoff on that BL, avoiding redundant write. If the cell needs to be written, the BL starts with high voltage. Once the write is done, the BL voltage drops and triggers the SA flipping to stop the write, avoiding overwrite to save the power. To ensure successful write, there are two ways to apply margin for write operation: 1) gate delay in self-termination control and 2) lower than threshold reference voltage.

V. MEASUREMENT RESULTS

The proposed 1-Mb STT-MRAM was fabricated in 28-nm embedded MRAM technology. Fig. 16 shows the die photograph of the MRAM chip. The 1-Mb MRAM macro occupies an area of 0.214 mm². It consists of eight arrays, each with 128-kb 1T1MTJ cells.

Fig. 17 shows the read failure rate across access time for a single 128-kb array. All error rates are reported without any error correction. The external V_{ref} keeps constant across temperature. With the self-generated V_{ref} , the array achieves 2.8- and 3.6-ns read access time with an approximately 1E-5 error rate at 25 °C and 120 °C, respectively. For access times less than 2.8 ns, higher error rates are observed with the self-generated V_{ref} because reference BLs require a longer stabilization time to drive the 16 sense amps. However, self-reference generation for read operation shows similar error rate as external V_{ref} with access times longer than 2.8 ns at room temperature but exhibits lower error rate at

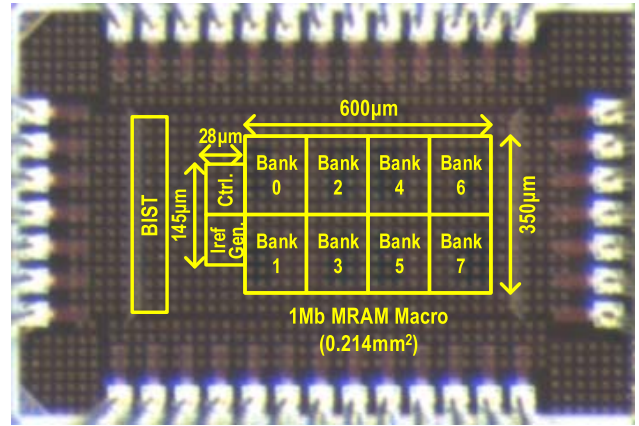


Fig. 16. Die photograph of 1-Mb STT-MRAM in 28-nm technology.

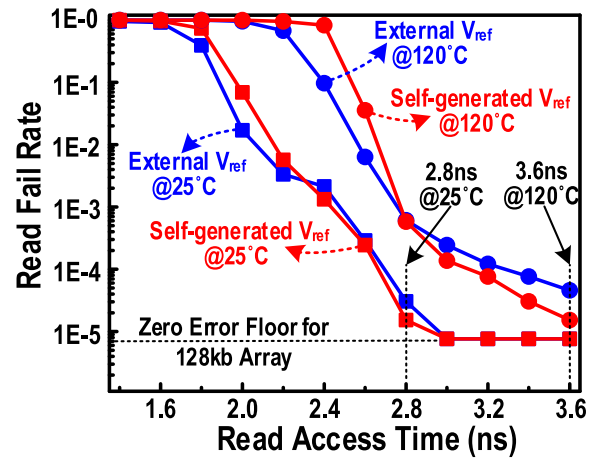


Fig. 17. Measured read failure rate across read access time for a single 128-kb array.

high temperature since it dynamically tracks with temperature. Fig. 18 shows the measured read fail rate across constant V_{ref} for eight arrays. We can see the optimal V_{ref} for each array could be different due to array-to-array variation which is caused by nonuniform CMP in the process or unbalanced voltage drop in each array. With a constant optimal V_{ref} across eight arrays, the blue circle shows the average error rate, which is 2E-5. The red circle illustrates the error rate with the self-generated V_{ref} . We can see that the self-generated V_{ref} tracks array-to-array variation, improving read failure rate by almost 2.2 times compared with a fixed external V_{ref} as measured across eight arrays.

Fig. 19 shows the read shmoo plot for the 1-Mb MRAM. At 1.2 V and 25 °C, 2.8-ns read access time can be achieved. Also, the memory can be successfully read at less than 0.6-V power supply. Fig. 20 shows the measured V_{DD_min} across 10 dies for the proposed sense amplifier. Due to offset cancellation, the sense amplifier is robust at low supply voltage with an average V_{DD_min} of 0.57 V and the standard deviation of 19 mV. For the read disturbance, even though the read current is as high as 1/4 of write current, the <3-ns read time is sufficiently short to avoid disturbance. According to the

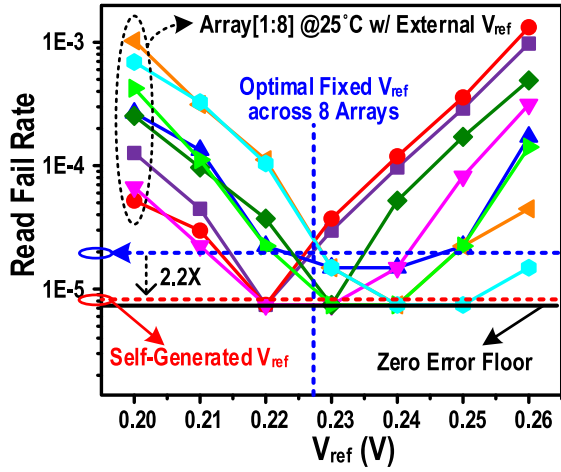


Fig. 18. Measured read failure rate across reference voltage for eight arrays.

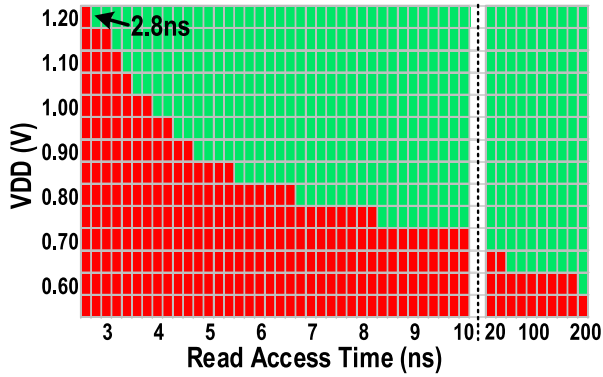
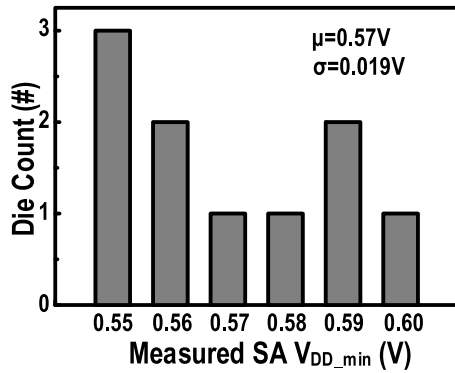


Fig. 19. Measured read shmoo plot.

Fig. 20. Measured V_{DD_min} across 10 dies.

measurement results, the read operation can be performed over $1E6$ times continuously without any disturbance observed.

Fig. 21 shows the write fail rate for 1 Mb and power saving ratio across write access time. The required write access time must be longer than 20 ns in order to achieve $1E-5$ failure rate using a conventional, fixed write time. Since the required write time varies significantly across cells, a constant write time wastes substantial write power and the proposed self-write-termination saves 47% and 61% write power for $1E-5$ and $1E-6$ write error rates (20 and 30 ns), respectively. When

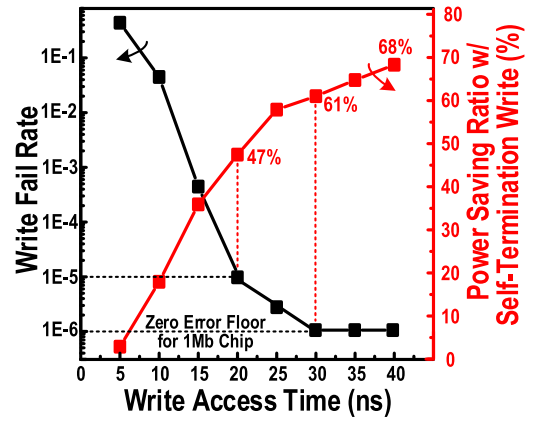


Fig. 21. Measured write fail rate for 1 Mb and power saving ratio across write access time.

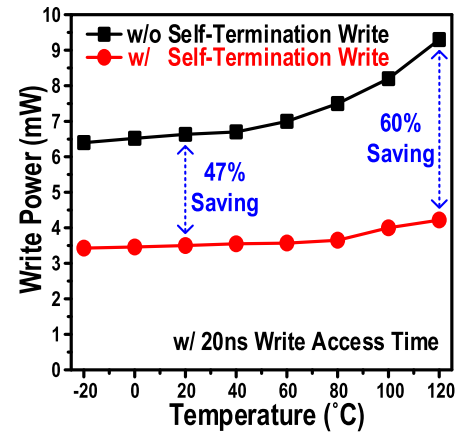


Fig. 22. Measured power consumption across temperature when writing checkerboard data.

TABLE I
COMPARISONS WITH OTHER WORKS

	This Work	ISSCC'13 [6]	ISSCC'10 [7]	ISSCC'10 [8]	ISSCC'09 [9]	ISSCC'15 [10]	VLSI'13 [11]
Technology (nm)	28	40	65	130	90	65	65
Cell Type	1T1MTJ	1T1MTJ	1T1MTJ	2T1MTJ	2T1MTJ	2T2MTJ	2T2MTJ
Target Application	NVM	NVM	NVM		LLC ¹⁾	LLC ¹⁾	LLC ¹⁾
Cell Area (F ²)	75		85	327	169		107
Capacity	1Mb	1Mb	64Mb	16kb	32Mb	1Mb	1Mb
Macro Area (mm ²)	0.214	0.57	47.124		91.02	0.8196	0.628
Power Supply (V)	1.2/1.8	1.1/2.5	1.2	1.2/3.3	1.5	1.2/0.9/0.4	1.2/0.9/0.4
Word Length (bit)	16	32	16		32	256	256
Read Speed (ns)	2.8	10	30	8	12	3.3	4
Write Speed (ns)	20		30	10	12	3	4
Read Power (mW)	3.9		7.8		60	21.6	17.8
Write Power (mW)	3.6		9.3	20	91	55.4	46.5
Read Energy (pJ/bit)	0.7		14.6		22.5	0.3	0.3
Write Energy (pJ/bit)	4.5		17.4		34.1	0.6	0.7

1) MTJ cell in LLC applications do not require high retention, allowing low switching current MTJs

the temperature increases, the self-write-termination is more effective (Fig. 22) because the required write time shortens at high temperature.

Table I shows the comparison of this paper to other MRAM works. Compared with the listed references, the proposed

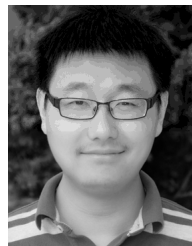
MRAM achieves best read access time and power consumption with the smallest macro area.

VI. CONCLUSION

A 1-Mb 28-nm 1T1MTJ embedded STT-MRAM is presented. Single-cap-based offset-cancelled sense amplifier is proposed to improve sensing margin and the self-reference generation method is used to track and mitigate the row-wise PVT variation. The 2.8-ns read access time and 0.57-V V_{DDmin} are achieved. Moreover, *in situ* self-write-termination is applied to avoid redundant write and auto terminate the write once write completes. Measured write power is reduced by 47% with 20-ns write access time.

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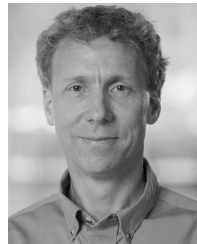
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